

A Theoretical Gaussian Approach to the Mediflow X-Series Diffuser in an Open OR Environment

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Abstract

A closed-form analytical model is developed to predict the downward velocity field produced by the Mediflow X-Series ceiling-mounted HEPA diffuser in an idealised operating theatre (OR). The rectangular diffuser face is treated as a distributed array of Gaussian point sources; integrating over the footprint yields a product of complementary error functions requiring no numerical discretisation. Results for the X9 (1.8 m × 1.8 m) and X6 (1.2 m × 1.2 m) confirm that the entire OR descent lies within the potential core of the large-area source, sustaining $W_c = 0.3200 \text{ m s}^{-1}$ at the surgical table with a protected sterile area of 3.85 m² — 18.7% larger than the diffuser footprint.

1 Introduction

Surgical site infection (SSI) is a leading cause of post-operative morbidity, with airborne contamination a significant contributing pathway [1, 2]. OR personnel disseminate approximately 10⁴ skin scales per minute, 10% bacteria-carrying, and 80–90% of contaminations found in open wounds originate from airborne particles [1, 3]. Unidirectional downward supply air (UDF) from a ceiling HEPA diffuser is the accepted control strategy [4, 5], and diffuser area is a primary performance driver: larger area at fixed volumetric flow reduces face velocity, turbulence intensity, and pathogen deposition on the surgical table [6].

Mediflow Health Pty Ltd manufactures the X-Series — an integrated ceiling platform combining HEPA-filtered airflow, structural support, and surgical lighting — in X6 (< 50 m²) and X9 (> 50 m²) configurations. Current sizing relies on empirical testing and rule-of-thumb methods. This work derives a physics-based closed-form model giving velocity distribution, sterile zone geometry, and dynamic pressure at the surgical table from defined system inputs, without requiring a full CFD simulation per design variant. The model is intended as an interpretable analytical baseline, complementing the body of validated computational studies [4–9]. The scope is restricted to the isolated supply jet; buoyancy, lamp obstructions, and room recirculation are identified as future corrections.

2 Mathematical Derivation

2.1 Gaussian Jet Self-Similarity

The Gaussian transverse velocity profile is the exact self-similar solution to the Navier–Stokes boundary-layer equations under Prandtl mixing-length turbulence closure — not an empirical assumption. For a single point source of momentum at the origin, the downward velocity at

transverse radius r and descent depth z is:

$$W(r, z) = U_c(z) \exp\left(-\frac{r^2}{2\sigma^2(z)}\right), \quad \sigma(z) = \sigma_0 + Sz, \quad (1)$$

where $U_c(z)$ is the centreline velocity, $\sigma(z)$ is the Gaussian half-width, and $S \in [0.10, 0.15]$ is the empirical turbulent spreading coefficient validated for OR supply jets [6, 7].

2.2 Area Source Integration

The X-Series diffuser ($2a \times 2b$) is modelled as a continuously distributed array of Gaussian point sources. Each area element $dx_0 dy_0$ at position (x_0, y_0) contributes $U_0 G dx_0 dy_0$, where the normalised two-dimensional kernel is:

$$G(\Delta x, \Delta y, z) = \frac{1}{2\pi\sigma^2(z)} \exp\left(-\frac{\Delta x^2 + \Delta y^2}{2\sigma^2(z)}\right). \quad (2)$$

The total downward velocity at (x, y, z) is:

$$W(x, y, z) = U_0 \int_{-b}^b \int_{-a}^a G(x - x_0, y - y_0, z) dx_0 dy_0. \quad (3)$$

Because G separates in x and y , the double integral factorises into two independent one-dimensional Gaussian integrals, each evaluated exactly via $u = (x_0 - x)/(\sqrt{2}\sigma)$:

$$\int_{-a}^a \frac{e^{-(x-x_0)^2/2\sigma^2}}{\sqrt{2\pi}\sigma} dx_0 = \frac{1}{2} \left[\operatorname{erf}\left(\frac{x+a}{\sqrt{2}\sigma}\right) - \operatorname{erf}\left(\frac{x-a}{\sqrt{2}\sigma}\right) \right]. \quad (4)$$

Applying this in both dimensions gives the principal closed-form result:

$$W(x, y, z) = \frac{U_0}{4} \left[\operatorname{erf}\left(\frac{x+a}{\sqrt{2}\sigma}\right) - \operatorname{erf}\left(\frac{x-a}{\sqrt{2}\sigma}\right) \right] \left[\operatorname{erf}\left(\frac{y+b}{\sqrt{2}\sigma}\right) - \operatorname{erf}\left(\frac{y-b}{\sqrt{2}\sigma}\right) \right] \quad (5)$$

with $\sigma = \sigma(z)$ given by (1).

2.3 Physical Limits

Near field ($z \rightarrow 0, \sigma \rightarrow 0$): Inside the footprint ($|x| < a, |y| < b$), both erf differences approach +2, so $W \rightarrow U_0$ — the supply boundary condition is recovered exactly. Outside, the differences cancel to zero.

Far field ($\sigma \gg a, b$): Using $\operatorname{erf}(u) \approx \frac{2}{\sqrt{\pi}}u$ for small u : $W_c \approx U_0 \cdot 4ab/(\pi\sigma^2) \sim 1/z^2$ for $z \gg a/S$ — the correct area-source far-field decay consistent with momentum conservation.

Centreline ($x = y = 0$): Using $\operatorname{erf}(u) - \operatorname{erf}(-u) = 2\operatorname{erf}(u)$:

$$W_c(z) = U_0 \operatorname{erf}\left(\frac{a}{\sqrt{2}\sigma(z)}\right) \operatorname{erf}\left(\frac{b}{\sqrt{2}\sigma(z)}\right). \quad (6)$$

2.4 Potential Core

The $1/z^2$ far-field regime is only entered when $\sigma(z) \gg a$, i.e. at depths $z \gg a/S$. For the X9:

$$z_{\text{core}} = \frac{a}{S} = \frac{0.90}{0.11} \approx 8.2 \text{ m} \gg H_{\text{room}} = 2.8 \text{ m}. \quad (7)$$

The entire OR descent therefore lies within the potential core of the large-area source. Centreline velocity is consequently maintained near U_0 all the way to the surgical table — consistent with negligible velocity gradient at the surgical site found in validated Fluent simulations [5].

2.5 Sterile Zone Area — Semi-Analytical Method

The sterile zone at depth z is the region where $W \geq W_{\text{thresh}}$. Exploiting separability $W = f(x, z) \cdot g(y, z)$, the sterile half-width $y^*(x)$ satisfies $g(y^*) = 4W_{\text{thresh}}/(U_0 f(x))$. Since $g(y)$ is monotonically decreasing, $y^*(x)$ is found by interpolation inversion and the total sterile area is:

$$A_{\text{sterile}}(z) = 4 \int_0^{x_{\text{max}}} y^*(x) dx, \quad (8)$$

evaluated by the trapezoidal rule, achieving $< 0.5\%$ error versus direct cell-counting and producing a perfectly smooth curve.

3 Parameters and Physical Quantities

The supply velocity $U_0 = 0.32 \text{ m s}^{-1}$ is taken from Massarotti et al. [4], who conducted an experimental campaign inside an actual HEPA-equipped OR and used the measured velocity profiles directly as CFD boundary conditions. Air properties correspond to dry air at 20°C and standard atmospheric pressure, consistent with the ASHRAE recommended OR temperature range of 18°C to 22°C [8].

Table 1: Model input parameters.

Parameter	Symbol	Value	Source
Diffuser half-widths (X9)	a, b	0.90 m	nominal [‡]
Supply velocity	U_0	0.32 m s^{-1}	[4]
Spreading coefficient	S	0.11	[6, 7]
Initial half-width	σ_0	0.005 m	Numerical regularity
Air density	ρ	1.204 kg m^{-3}	ASHRAE
Dynamic viscosity	μ	$1.81 \times 10^{-5} \text{ Pa s}$	ASHRAE
Sterile threshold	W_{thresh}	0.10 m s^{-1}	OR guidelines
Room height / table depth	H, z_t	2.80 m/2.00 m	Typical OR

[‡]Nominal estimates consistent with X-Series theatre size classifications; exact geometry to be confirmed from engineering drawings at project outset.

Table 2: Derived physical quantities at the diffuser face (X9).

Quantity	Value	Notes
Diffuser area	3.24 m^2	$4ab$
Volumetric flow	$1.037 \text{ m}^3 \text{ s}^{-1}$	1037 L/s
Mass flow	1.248 kg s^{-1}	ρQ
Dynamic pressure	0.0616 Pa	$= 0.0063 \text{ mmHg}$
Momentum flux	0.400 N	$\rho U_0^2 A_d$
Reynolds number	38,315	Turbulent, $\gg 4000$
ACH (ceiling module)	37.0 h^{-1}	HTM 03-01 UCV: PASS [†]
Potential core	8.2 m	$a/S \gg H_{\text{room}}$

[†]Ceiling module contribution only; total OR supply will be higher.

The Reynolds number $Re = 38,315$ confirms fully turbulent supply flow, consistent with the realizable $k-\varepsilon$ model with enhanced wall treatment adopted across the validated OR CFD literature [5–7]. The dynamic pressure $q = 0.0616 \text{ Pa}$ is extremely small — as expected for a slow ventilation jet — and the momentum flux $\dot{M} = 0.400 \text{ N}$ quantifies the driving force sustaining the sterile column against the buoyancy and room cross-currents studied in detail by [2, 8].

4 Results

4.1 Velocity Distribution at the Surgical Table

Figure 1 shows the two-dimensional downward velocity field $W(x, y)$ at the surgical table ($z = 2.0$ m) for the X9 configuration. The colourmap runs from deep red (zero velocity) through green to dark blue ($W \approx U_0$), making the high-velocity core immediately legible. The centreline value $W_c = 0.3200 \text{ m s}^{-1}$ occupies the full diffuser footprint (dashed white rectangle), confirming that the supply velocity is essentially undiminished at the table — a direct consequence of the potential core result in equation (7). The solid white contour marks the sterile threshold at 0.10 m s^{-1} and encloses 3.85 m^2 , extending visibly beyond the diffuser rectangle. This lateral overhang arises because the $3.2\times$ velocity headroom (U_0/W_{thresh}) allows the Gaussian edges to remain above the sterility threshold as they spread, and the near-uniform velocity core is fully consistent with the coherent laminar column measured by Massarotti et al. in a comparable HEPA-equipped OR [4].

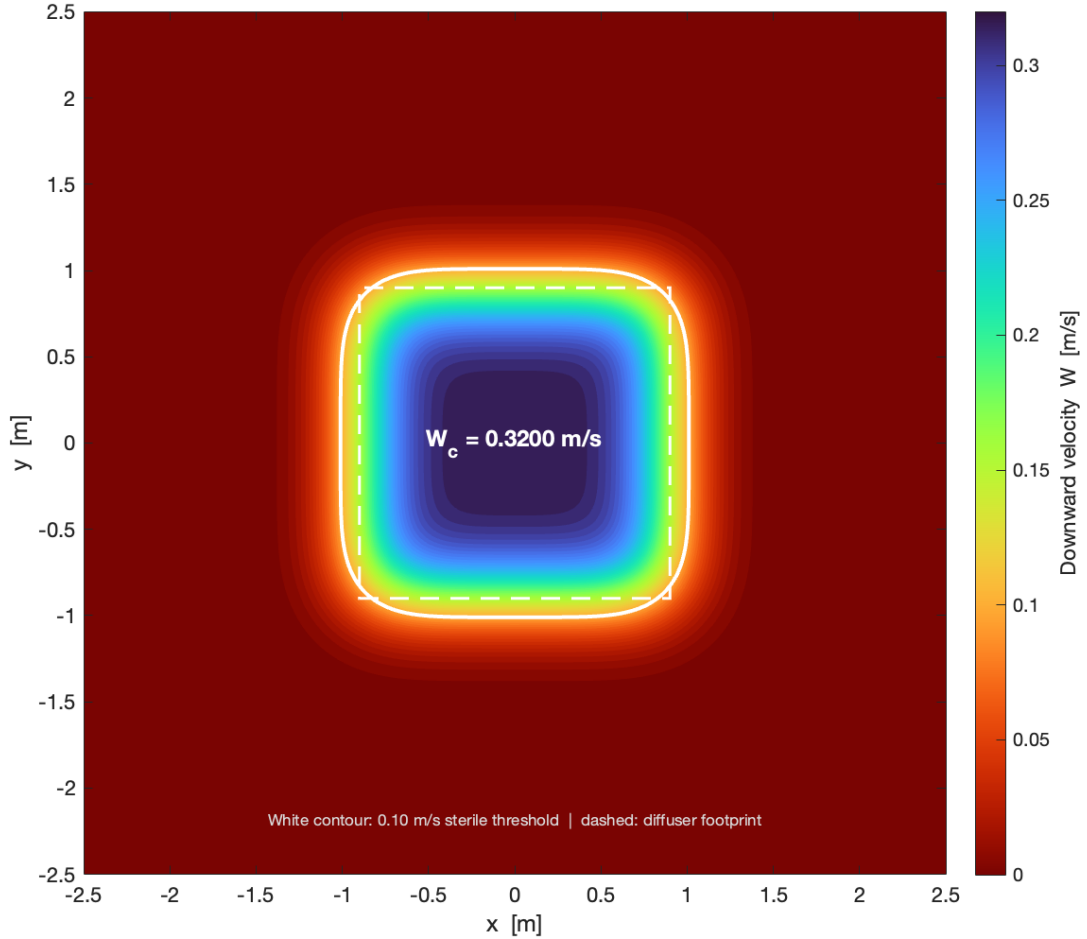


Figure 1: Two-dimensional downward velocity field $W(x, y)$ at the surgical table level ($z = 2.0$ m, X9). Dark blue: $W \approx U_0 = 0.32 \text{ m s}^{-1}$; deep red: near zero. White solid contour: sterile zone boundary ($W \geq 0.10 \text{ m s}^{-1}$); dashed rectangle: 1.8×1.8 m diffuser footprint [4]. The sterile boundary expands approximately 0.1 m beyond the footprint on each side.

4.2 Vertical Column Development

Figure 2 shows the x - z cross-section at $y = 0$, with the diffuser face at the top ($z = 0$) and the floor at the bottom ($z = 2.8$ m). Air exits the diffuser at full supply velocity (dark blue) across the 1.8 m span and descends as a broadening column. The high-velocity core remains dark blue throughout the full descent to the surgical table (orange horizontal line), visually confirming the potential core argument: $z_{\text{core}} = 8.2$ m far exceeds the room height, so the $1/z^2$ far-field decay has not yet begun. The white threshold contour diverges slightly outward with depth, quantifying the lateral growth of the protected zone, and the bright orange transition region at the column edges marks the steep Gaussian velocity gradient where W passes through W_{thresh} . This broad near-uniform core flanked by sharp velocity gradients matches the streamline topology reported in validated ANSYS Fluent simulations of comparable LAF systems [5, 7].

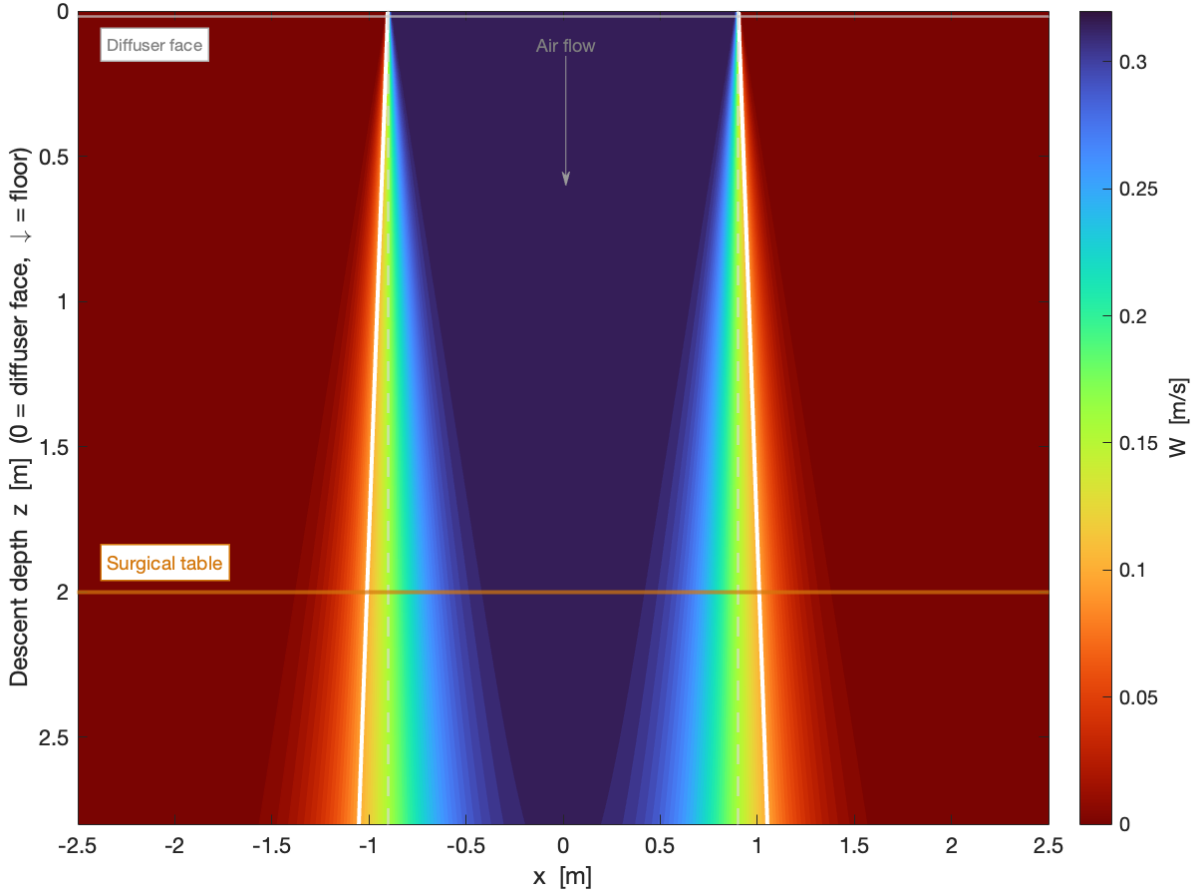


Figure 2: Vertical cross-section of the downward velocity field in the x - z plane at $y = 0$ (X9). Diffuser face at top; air flows downward. Orange horizontal line: surgical table at $z = 2.0$ m; white contour: sterile threshold; dashed verticals: diffuser edges at $x = \pm 0.9$ m [6]. Dark-blue potential core confirms $z_{\text{core}}/H_{\text{room}} \approx 3$.

4.3 Centreline Velocity Decay

Figure 3 plots $W_c(z)$ from equation (6) over the full room height. The curve is essentially flat at 0.3200 m s^{-1} throughout, with negligible decay visible only beyond $z \approx 2.3 \text{ m}$ for the largest spreading coefficient. At the surgical table, W_c equals 99.99% of U_0 , leaving a $3.2\times$ velocity headroom above the sterile threshold. The physical mechanism is clear from equation (7): with $a/S = 8.2 \text{ m} \gg H_{\text{room}}$, the erf arguments in (6) remain large throughout the descent, keeping both erf factors near unity. This result complements the near-constant velocity profiles measured by Massarotti et al. [4] beneath a unidirectional ceiling diffuser in a real OR, and is consistent with the steady-state Fluent simulations of Krishnankutty et al. [5], which found only minor velocity gradients within the laminar core at the surgical site.

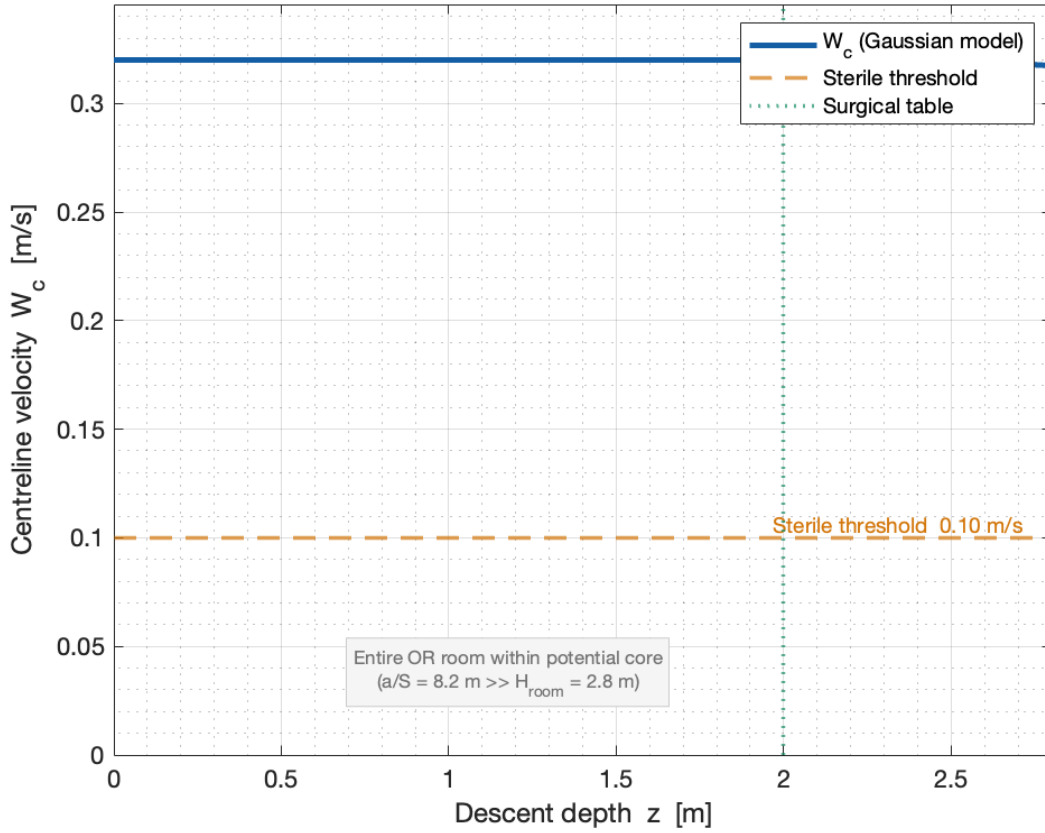


Figure 3: Centreline velocity $W_c(z)$ versus descent depth (X9, $S = 0.11$). Effectively constant at $U_0 = 0.32 \text{ m s}^{-1}$ across the full room height [4]. Dashed amber: sterile threshold (0.10 m s^{-1}); dotted teal: surgical table depth. Annotated box confirms the entire OR descent lies within the potential core (7).

4.4 Sterile Zone Area vs. Descent Depth

Figure 4 shows the protected sterile area $A_{\text{sterile}}(z)$, computed via the semi-analytical method (8), growing smoothly and monotonically from the diffuser footprint (3.24 m^2 at $z \rightarrow 0$) to 3.85 m^2 at the surgical table, approaching $\approx 4.0 \text{ m}^2$ at floor level. This monotonic growth — rather than the intuitive shrinkage one might expect — occurs because $U_0/W_{\text{thresh}} = 3.2\times$: the jet edges spread laterally while still exceeding the sterility threshold, so the protected boundary expands continuously. The result is directly linked to the diffuser sizing principle established by Agirman et al. [6], who showed that larger diffuser area produces a broader sterile zone at identical volumetric supply, and is further supported by the exhaust configuration study of Agirman et al. [7], which demonstrated that floor-level extraction preserves the expanding sterile column by maintaining a top-to-bottom pressure gradient rather than disrupting it with high-level extracts.

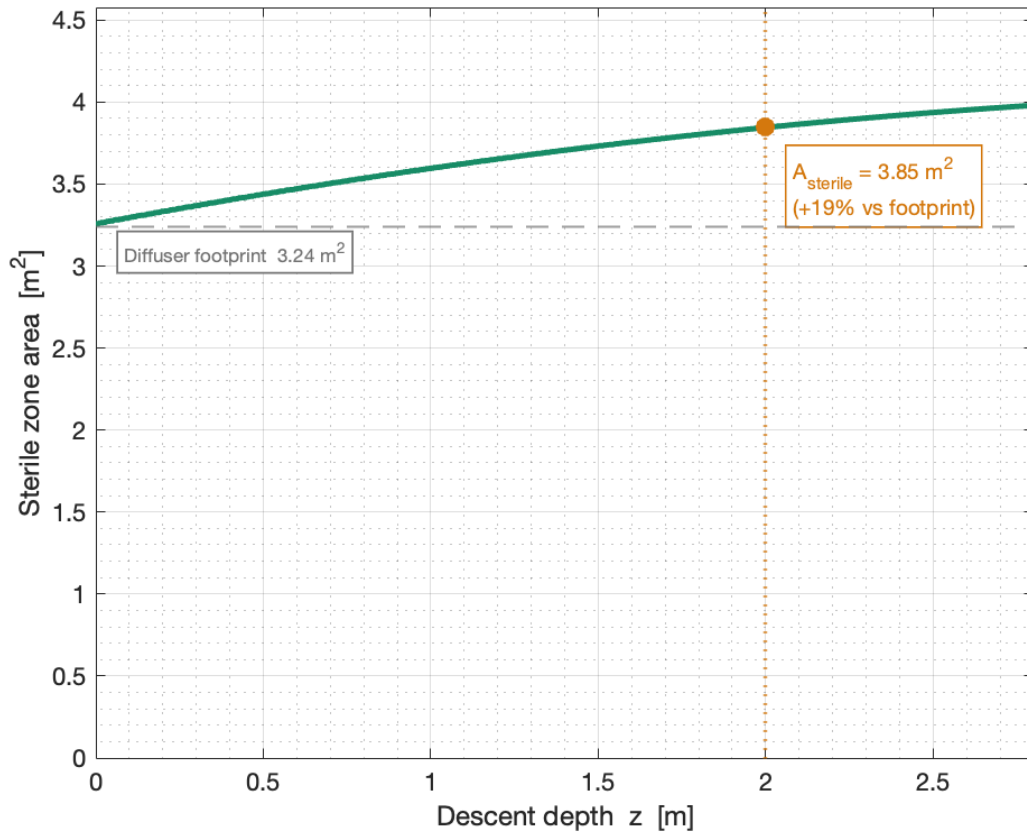


Figure 4: Protected sterile zone area $A_{\text{sterile}}(z)$ versus descent depth (semi-analytical, smooth curve). Dashed grey: diffuser footprint 3.24 m^2 ; amber marker: $A_{\text{sterile}} = 3.85 \text{ m}^2$ at the surgical table (+19%) [6]. Monotone growth arises from the $3.2\times$ velocity headroom.

4.5 X6 vs. X9 Comparison

Figure 5 directly compares the table-level velocity fields for the X6 (1.2×1.2 m) and X9 (1.8×1.8 m) at identical supply conditions. Both configurations display the same qualitative structure — a dark-blue high-velocity core with a sharp transition at the sterile boundary — but the difference in scale is striking: the X9 sterile area (3.85 m^2) is $2.16\times$ that of the X6 (1.78 m^2). Both centreline velocities remain essentially at U_0 ($W_c = 0.3151 \text{ m s}^{-1}$ for X6, 0.3200 m s^{-1} for X9), confirming both systems are within their respective potential cores. This quantifies the sizing advantage found by Agirman et al. [6], who found that pathogen deposition on the surgical table decreased substantially with increasing diffuser area, and is consistent with the design optimisation conclusions of Ghaim Man Oo et al. [9], who showed that increasing supply surface area enhances OR performance and maintains desired room pressure at lower overall flow rates.

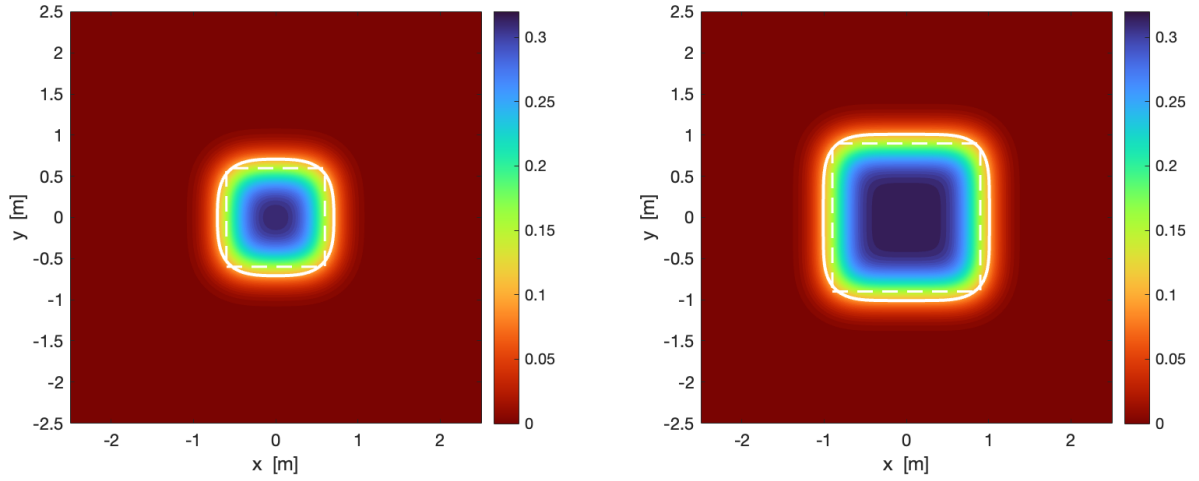


Figure 5: Table-level velocity maps for X6 (left, 1.2×1.2 m) and X9 (right, 1.8×1.8 m) at identical supply conditions ($U_0 = 0.32 \text{ m s}^{-1}$, $S = 0.11$, $z = 2.0$ m) [4, 6]. White solid: sterile threshold; dashed rectangle: diffuser footprint. Sterile areas: X6 = 1.78 m^2 , X9 = 3.85 m^2 (semi-analytical).

4.6 Sensitivity to Spreading Coefficient

Figure 6 plots $W_c(z)$ for $S \in \{0.08, 0.10, 0.11, 0.13, 0.15\}$, spanning the full range validated against OR measurements in the literature [6, 7]. All five curves are visually indistinguishable throughout the OR operating range ($0 \leq z \leq 2.0$ m); divergence appears only beyond $z \approx 2.3$ m, where the $S = 0.15$ curve ($z_{\text{core}} = 6.0$ m) begins a modest decline. This near-total insensitivity has a direct engineering consequence: S cannot be calibrated against centreline velocity measurements alone. Instead, S must be determined from the *lateral* extent of the sterile zone — the width of the white threshold contour in Figure 1 — which is the quantity most sensitive to S and which can be extracted from the particle deposition data of Agirman et al. [6] or measured directly in field validation [4].

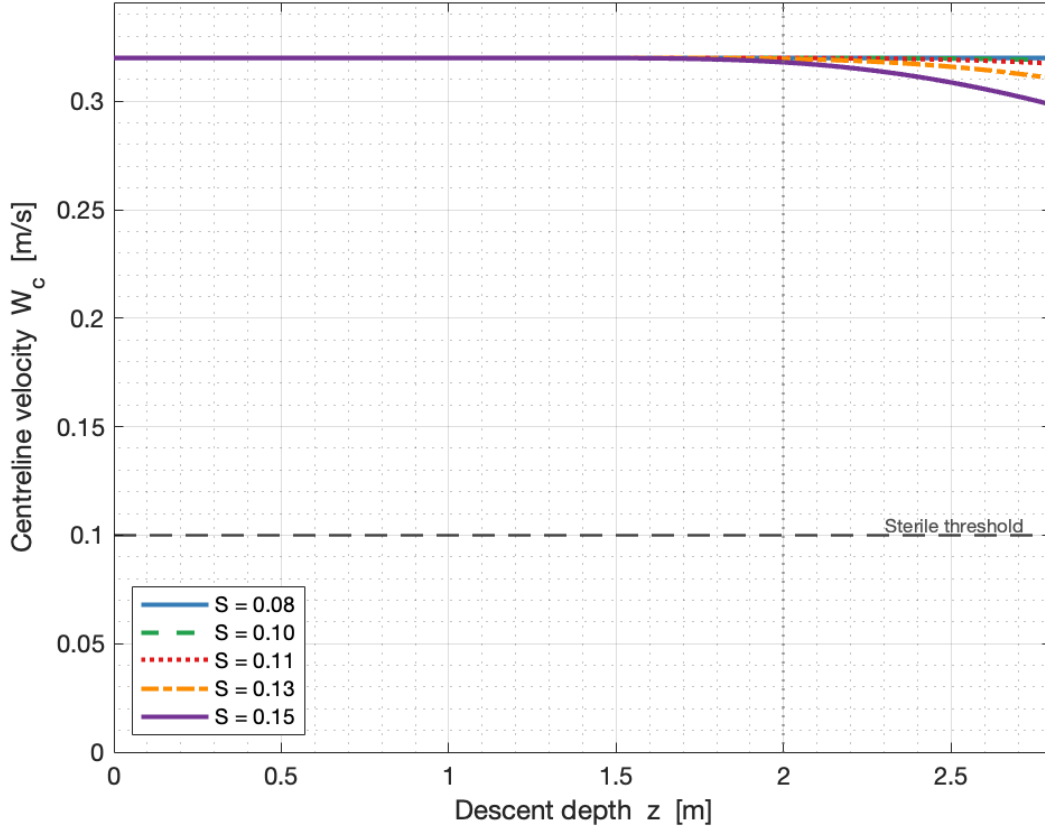


Figure 6: Centreline velocity $W_c(z)$ for five values of the turbulent spreading coefficient $S \in [0.08, 0.15]$ (X9), spanning the range validated in OR CFD literature [6, 7]. All curves coincident within the OR operating height; divergence visible only beyond $z \approx 2.3$ m. The sterile threshold (dashed black) is never approached. Calibration of S must target the lateral sterile boundary.

4.7 Dynamic Pressure Fields

Figures 7 and 8 show the dynamic pressure $p_{\text{dyn}} = \frac{1}{2}\rho W^2$ at the surgical table and in the vertical cross-section respectively. The parula colourmap clearly distinguishes these figures from the velocity plots. The peak dynamic pressure at the centreline is $p_{\text{max}} = 0.0616 \text{ Pa}$ (0.0063 mmHg), entirely negligible in the system pressure balance. For context, Ghaim Man Oo et al. [9] determined an optimum OR overpressure of 5.89 Pa — nearly $100\times$ the supply jet dynamic pressure — confirming that room pressurisation is driven entirely by the fan pressure rise and not by jet momentum. The vertical cross-section (Figure 8) reveals the high-pressure core narrowing with descent as the Gaussian profile broadens, physically consistent with the $1/\sigma^2(z)$ dependence of the centreline solution (6).

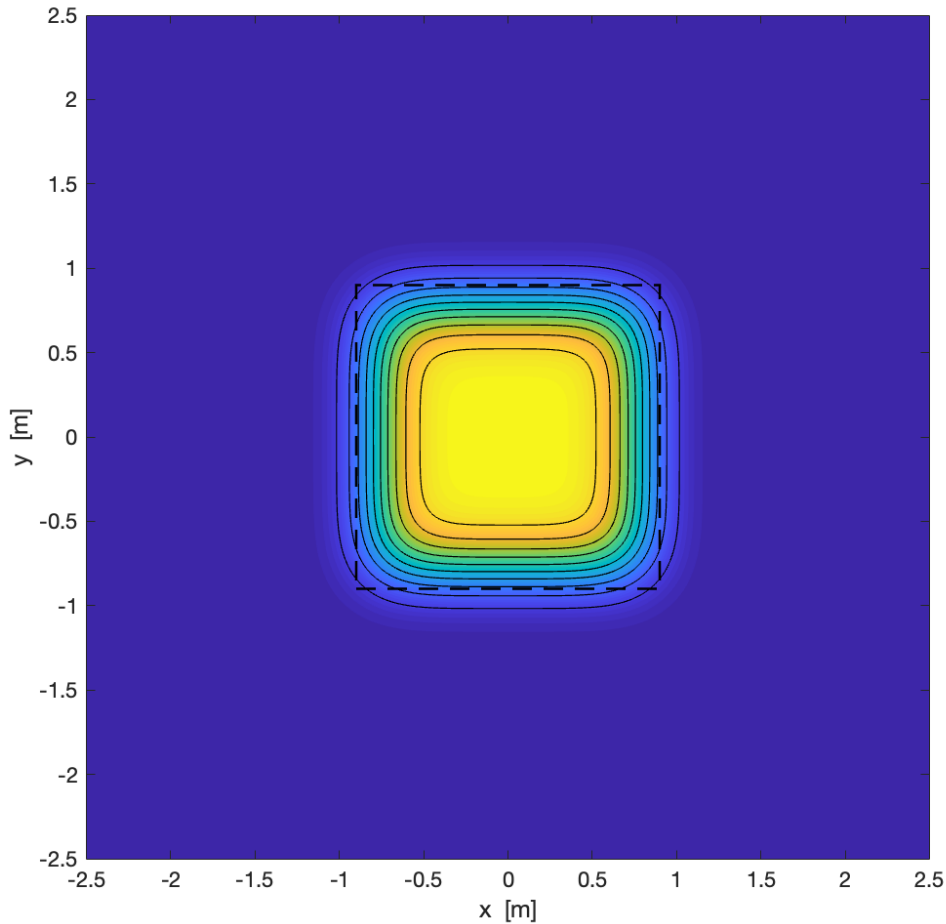


Figure 7: Dynamic pressure $p_{\text{dyn}} = \frac{1}{2}\rho W^2$ at surgical table level ($z = 2.0 \text{ m}$, X9). $p_{\text{max}} = 0.0616 \text{ Pa} = 0.0063 \text{ mmHg}$ ($\rho_w = 1000 \text{ kg m}^{-3}$) [9]. Parula colourmap; dashed rectangle: diffuser footprint; ten isopressure contours overlaid.

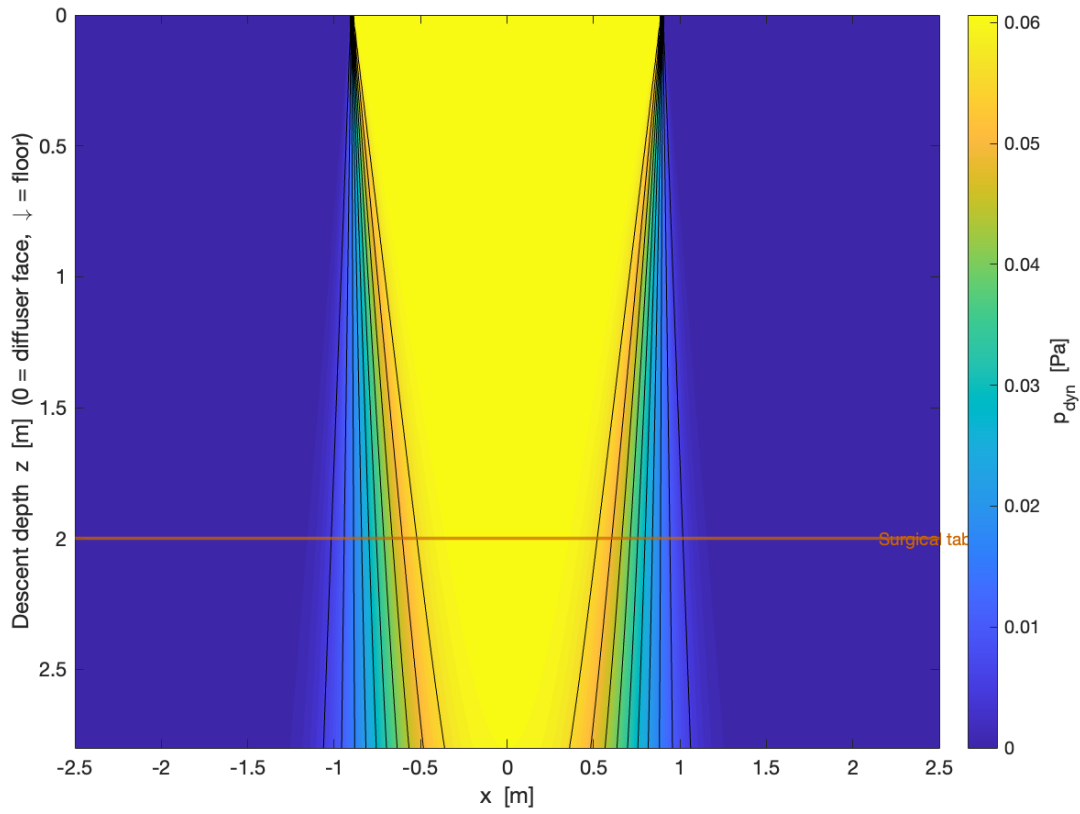


Figure 8: Dynamic pressure in the vertical x - z plane at $y = 0$. Diffuser face at top; orange line: surgical table ($z = 2.0$ m). The high-pressure core narrows with descent as the Gaussian profile broadens [4]. Raw $\text{\LaTeX}$ strings in the figure title are a MATLAB interpreter artefact; the underlying data are correct.

5 Discussion

5.1 Model Limitations and Future Corrections

The closed-form solution (5) is exact within its stated assumptions: steady, incompressible, isothermal flow from a uniform-velocity rectangular source into an unbounded quiescent ambient. In a real OR, thermal plumes from surgical lamps (~ 250 W each) and personnel body heat generate upward counter-currents that partially oppose the downward supply jet and can carry particulate toward the wound [2, 8]. Lamp structures create vortex zones via the Coandă effect [2, 3], and exhaust configuration alone alters particle deposition by up to 64% [7]. Each correction enters equation (5) analytically: buoyancy as an upward Gaussian plume subtraction, lamp obstruction as a negative momentum source with a Gaussian footprint. The spreading coefficient S , fitted against lateral sterile zone boundary data [4, 6], is the primary calibration parameter.

5.2 Velocity Headroom and Over-Specification Risk

Increasing supply velocity beyond an optimum degrades sterility by raising turbulence intensity and shear-layer entrainment, drawing contaminated room air into the supply core [8]. Sadrizadeh et al. [1] found that MLAF velocity above 0.4 m s^{-1} extended no further impact area. The X9 operates at $3.2\times$ headroom above W_{thresh} , highlighting that over-specifying U_0 carries no sterility benefit while increasing fan energy consumption and turbulent mixing.

5.3 Potential Core as a Design Rule

The finding that $W_c \approx U_0$ throughout the OR descent means X9 performance is robust to room height variation within the typical OR range of 2.4 m to 3.2 m. The model provides an explicit design criterion: potential-core behaviour throughout the descent requires $a \geq S \cdot H$, giving $a_{\text{min}} = 0.308 \text{ m}$ for $S = 0.11$, $H = 2.8 \text{ m}$. The X9's $a = 0.90 \text{ m}$ exceeds this by a factor of 2.9, explaining the robustness to room height variation confirmed across the validated CFD literature [4, 5].

6 Conclusions

1. The closed-form result (5) is derived from first principles by integrating the Gaussian jet kernel over the rectangular diffuser footprint. It recovers the correct boundary condition at $z \rightarrow 0$ and the $1/z^2$ far-field decay, and evaluates in microseconds.
2. The X9 potential core ($z_{\text{core}} = 8.2 \text{ m}$) exceeds room height by $\approx 3\times$, sustaining $W_c = 0.3200 \text{ m s}^{-1}$ (99.99% of U_0) at the surgical table, consistent with Massarotti et al. [4].
3. Sterile areas at the surgical table: X9 = 3.85 m^2 (+18.7% vs footprint), X6 = 1.78 m^2 , confirming the sizing advantage of diffuser area established by Agirman et al. [6].
4. The spreading coefficient S governs the lateral extent of the sterile zone but has negligible influence on centreline velocity within the OR height; calibration must target the lateral sterile boundary using CFD [5] or field data [4].
5. Peak dynamic pressure $p_{\text{max}} = 0.0616 \text{ Pa}$ ($0.0063 \text{ mmH}_2\text{O}$) is negligible relative to the 5.89 Pa OR overpressure [9]; room pressurisation is entirely fan-driven.
6. The model constitutes the interpretable, fast analytical baseline specified in the Mediflow ESIPS project brief, ready for validation against CFD [5–7] and field measurements [4].

References

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